Kinds of Electric Circuits: (3)

There are three basic kinds of electric circuits:

1. The Series Circuits.
2. The Parallel Circuits.
3. The Series-Parallel Circuits.

The Series circuits: In a series circuit, electricity has only one path in which to travel from beginning to end. The flashlight is an example of this kind of circuit.

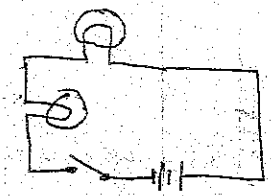


fig Series circuit.

Parallel Circuit: In parallel circuit, there are many paths or "branches" for the electricity to take. Some christmass-tree lights are good examples of parallel circuit.

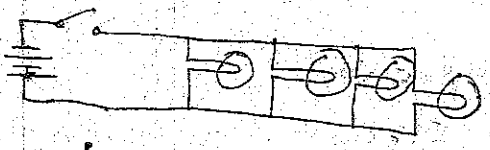
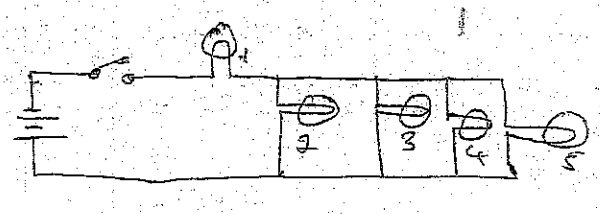


fig Parallel circuit.

if one light goes out because the filament breaks, the other bulbs will remain bright. This happens because the burned-out bulb no longer has a complete path but it does not affect the other branches. Your electric light at home are connected it parallel.

Series Parallel Circuits: This is a Combination of Series ~~Circuit~~ and parallel Circuit. from the fig below, Lamp 1 is in Series, while Lamp 2, 3 of 4, 5, make up of a parallel Circuit. Even the most advanced Circuit are made up of Series-parallel Circuits.



ELECTRONIC CIRCUITS

Electronic devices are devices that use electron tubes and transistors. Their circuits are called electronic Circuits. Radio and television sets are examples of electronic devices. Electronic circuits may look complicated, but they are usually only a Combination of many simple Circuits. Each Circuit can be traced seperately. You will find these Circuits Connected in Series or in Series-parallel.

Circuit Diagrams: Circuit diagrams show us where the different parts of a Circuit are located and how these parts are Connected. They are used when equipment is designed, ~~Constructed~~, maintained Constructed, maintained, serviced & repaired. Without such diagrams, many jobs would become Largely guesswork and Could not be done in a well planned manner. In electrical and electronic Work, three main kinds of Circuit diagrams are Used.

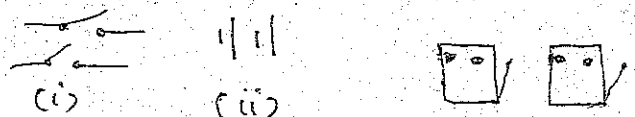
1. Pictorial diagrams
2. Schematic wiring diagrams
3. Floor-plan diagrams.

Pictorial Diagram: A pictorial diagrame is a "picture" showing the parts of a circuit, where they are located, and how they are connected. In such a diagram, the circuit is shown as it will look when finished. Because they are easy to follow and to understand, pictorial diagrams are often used when new equipment is being constructed. The diagrams are furnished with the "do it your self"

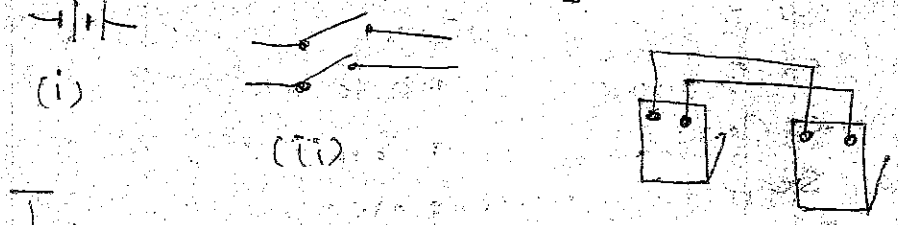
Schematic Wiring diagram: Some electric circuits have many parts. Given either a cut away or a pictorial view of each of these parts would take a great deal of time and material. Because a workers time cost money, a simplified drawing called a schematic wiring diagram is used. A schematic wiring diagrams uses symbols in place of a cutaway view of an object. The schematic wiring diagram saves the time of the draftman and is easy to read once you know the meaning of the symbols. When you draw a schematic wiring diagram of a circuit, you must know the parts that are going to be used in the circuit. You must also know the purpose for which the circuit is to be used.

The following procedure is recommended for drawing a schematic diagram of a circuit

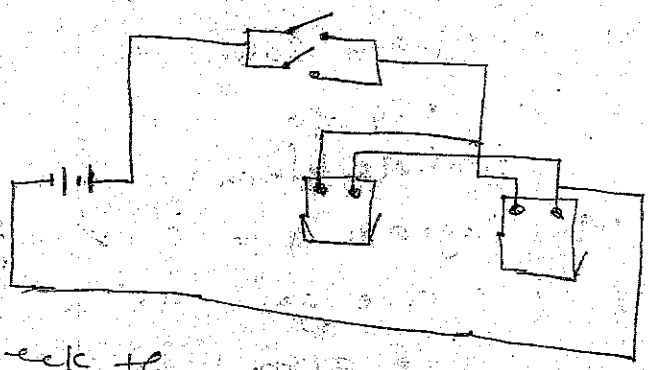
1. Place the parts' symbols on a sheet of paper, locating all like symbols on a sheet of paper, locating all like symbols near each other e.g.



2. Connect the symbols. You do not have to draw a line between dry cells connected in series. Simply place the negative terminal of one cell next to the positive terminal of the other cell. To connect the switches and buzzers in parallel, draw a line joining terminals in the same position; left to left right to right, upper to upper, and so on. e.g



3. Join all these parts to form a complete circuit



4. Check the diagram and see that all parts are have been connected properly.

Some areas in which capacitors are used includes:

i. It is used in charge pump circuit as energy storage device.

ii. Capacitors are used for power conditioning, signal coupling or decoupling.

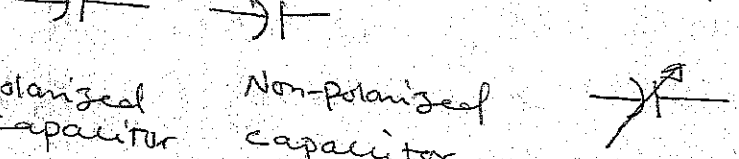


fig K. Polarized capacitor, Non-polarized capacitor, Variable capacitor

L. Transformer: Transformer is an electrical device which transforms voltage level from one stage to another.

ROTATING ELECTRICAL MACHINES:

The general symbol for an electrical ~~machine~~ machine is a circle which encloses full information denoting the type of machine. This includes whether it is a generator (G) or a motor (M) or is capable of alternate use as a motor or generator (M or G). Further information included is whether the machine is designed for direct or alternating supply. If d.c. a short straight line is placed below the letter designation if a.c. a small representation of sine wave is placed below the letter.

Symbol for machines for alternating supply may indicate the number of phase by number or in some cases by a symbol if the type three-phase connection is specified. Induction machines have a second circle within the main one.

The field windings of d.c. machines may be classed as shunt fields, series field and commutating or compensating fields and are represented as in the diagrams below; where machines are to be mechanically coupled, two straight lines are drawn between each circle to represent the coupling as in figure below (fig m).

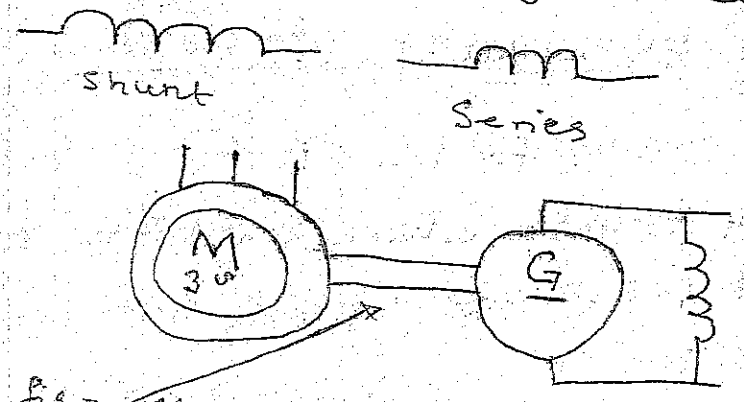
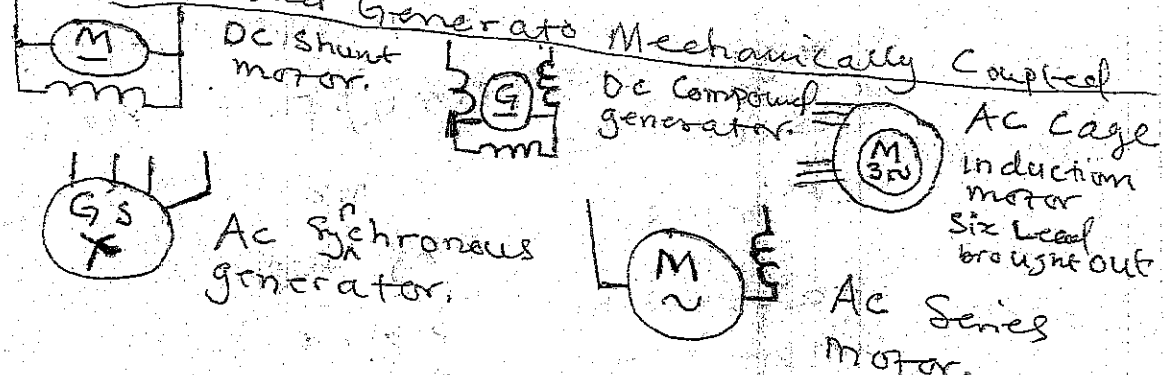
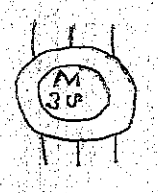
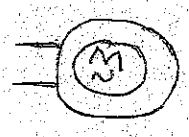


fig z — Motor and Generator Mechanically Coupled





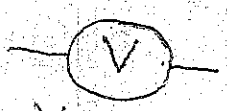
Ac wound Rotor Induction Motor.



Ac Single phase Induction Motor.

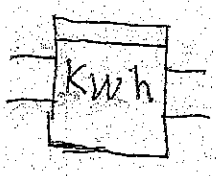
Indicating and Recording Instruments.

An indicating instrument is drawn as a circle with a letter within the circle signifying the type. A recording instrument is represented by a square and for an intergrating instrument such as a kilowatt-hour meter, a small rectangle is added to the square. Only a selection of instruments will be illustrated here

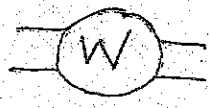


Voltmeter.

Ammeter.



Kilowatt-hour meter.



Wattmeter.

ELECTROMAGNETICALLY OPERATED SWITCHES:

Switches may be operated manually or may be actuated by a particular method. Eg. Mechanically, electromagnetically, manually etc. Electromagnetically operated switches are either called relays or contactors. There is a little difference between them, only that relay usually refers to a small current carrying electromagnetic switch which which in turn operates a high current carry electromagnetic switch called a Contactor

Contactors may be operated directly without a relay.

NB: for more about switches, go through the diagrams given to you.

What is a switch?

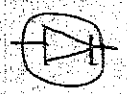
A switch is simply a device which is made to make or open and electrical/electronic device

Review Questions

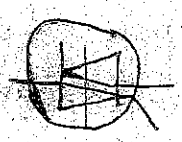
1. Why do you think, it is important to understand the Language of Electricity
2. Briefly discribe with the aid of a diagrams the following and give one area in which each is used.
   i. Toggle switch
   ii. Cell
   iii. Battery
   iv. Electric Bell and Buzzers
   v. Electrical motor
   vi. Resistor
   vii. Transistor
   viii. Diode
3. Differentiate betwee Transformer and Inductor
4. Draw the electrical symbol of the following
   i. Dc shunt motor, ii Ac Synchronous generator
   iii. Ac Cage Induction motor, iv. Ac Serice motor
   v. Ac Synchronous generator.
5. Briefly discribe what you understand by Indicating and Recording Instrument

SEMICONDUCTORS (5)

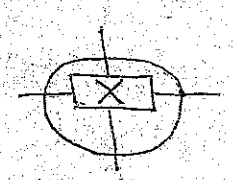
Semiconductors belongs to a general class of material that are neither good Conductors nor good Insulators. Under Some Condition they act as Conductor Under Some Conditions they act as Insulators. Three of the most Common Semiconductor Materials are germanium, Silicon, and Selenium. These materials are used in making transistors, rectifier diodes, and other Solid State devices. Below, are Some example of Semiconductor devices:



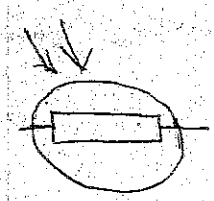
Diode



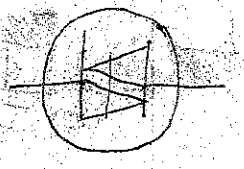
Triac



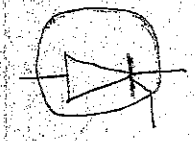
Hall Generator



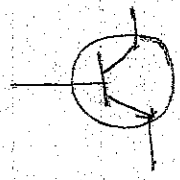
Light dependant Resistor (LDR)



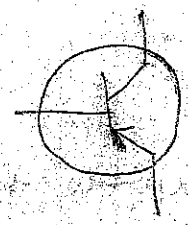
DIAC



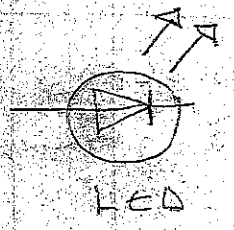
SCR



NPN BJT

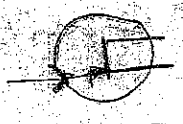


PNP BJT



LED

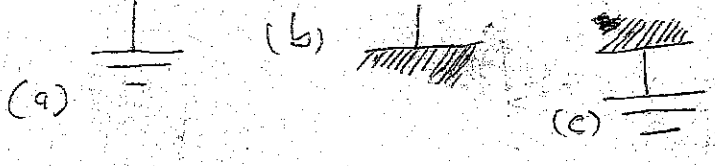
UJT



FET

EARTH CONNECTIONS:

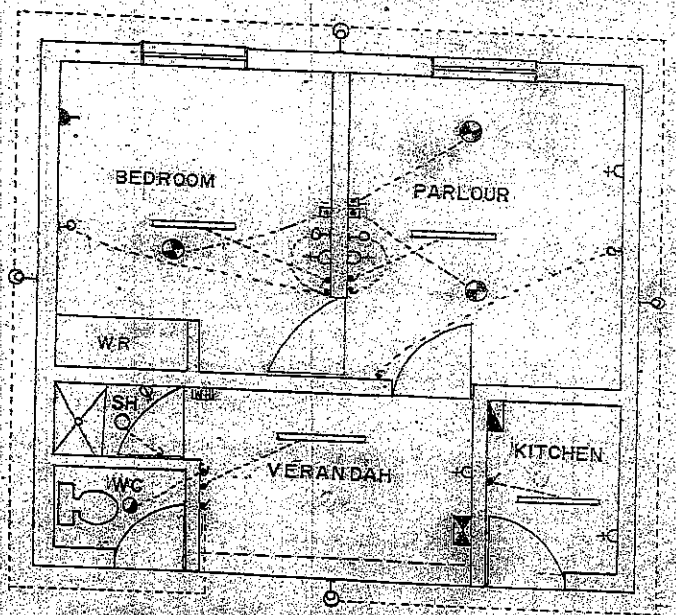
The term earthing means to electrically connect conductors or apparatus to the general mass of earth, but may at times be apply to connection to the metalic frame or chassis of an appliance. To distinguish between these two similar applications, different symbol have been employed. In the figure below, the two symbols are illustrated together with a combination of the two which signifies the connection of the metalic frame of the electrical apparatus to the general mass of earth.



Review Questions:

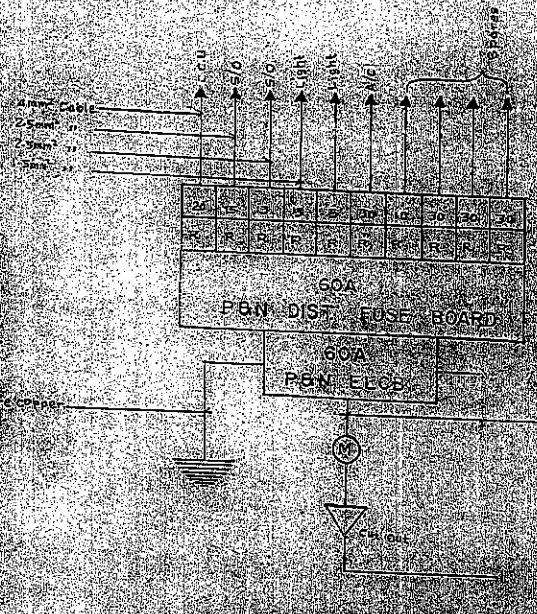
1. Discribe briefly with the aid of a diagram, the term Earth Connection.
2. Draw any ten (10) Semiconductor devices you know.

DO NOT PHOTOCOPY



FLOOR PLAN (Electrical Installation)

LEGEND	
	Distribution fuse board
	Cooker control unit
	Wall bracket
	Security lights
	Flourescent fittings
	Water heater points.
WH	Water heater switch
	Ceiling fan point
	Fan regulator
	15A.S.S.O.
	13A.S.S.O.
	Switches
	Ceiling mounted waterlight Screw neck fitting



ONE BEDROOM AND PARLOUR	
Designed by	I. M. Ibrahim
Drawn by	I.M. Ibrahim
Checked by	A. D. Dzoro
Vetted by	A. D. Dzoro
Approved by	A. D. Dzoro
Scale	1:50
Date	07-01-1997